

# Module 06: Learning in Robotics — Autonomous Lifelong Learning

## Learning Objectives

1. Define **lifelong learning** as the autonomous agent's capacity to continuously improve its generative model over extended deployment without human retraining.
2. Analyze the challenges of **catastrophic forgetting, distributional shift, and self-supervised learning** in autonomous robotic systems.
3. Apply Active Inference learning mechanisms to enable robots that improve with experience.

## Introduction

A truly autonomous robot must learn throughout its operational lifetime — adapting to new environments, new tasks, and changing conditions without being taken offline for retraining. This is **lifelong learning**: the continuous improvement of the generative model from the robot's own experience, balanced against the risk of forgetting previously learned competencies.

## Key Concepts

### 1. Self-Supervised Learning from Experience

Autonomous robots generate their own training data through operation:

- Every action the robot takes produces observed consequences — a natural source of (state, action, next state) supervision for the B matrix
- Every observation the robot makes provides evidence for the A matrix — the robot continuously calibrates its sensor models
- **Prediction error as learning signal**: Moments of high prediction error are the most informative for learning — they indicate where the current model is wrong

Self-supervised learning is Active Inference learning in its purest form: the agent uses its own prediction errors as the signal for model improvement.

## 2. Catastrophic Forgetting

**Catastrophic forgetting** occurs when learning new information destroys previously learned knowledge:

- A robot that learns a new room layout may degrade its model of previously learned rooms
- A manipulation robot that adapts to a new tool may lose proficiency with the old tool
- Active Inference mitigates forgetting through **concentration parameter accumulation** — well-learned knowledge (high concentration parameters) is resistant to overwriting, while new knowledge updates parameters with lower concentrations

## 3. Distributional Shift

The environment changes over time, causing **distributional shift**:

- Seasonal changes (lighting, temperature, vegetation) shift visual observations
- Infrastructure changes (construction, renovation) shift spatial models
- Personnel changes (new staff, different schedules) shift social models

Autonomous learners must distinguish between model error (the model was always wrong) and distributional shift (the model was right but the world changed). Active Inference handles this through precision monitoring — sudden increases in prediction error with high prior precision suggest distributional shift rather than ordinary model error.

## 4. Curriculum Learning and Self-Assessment

Autonomous learners can manage their own learning curriculum:

- **Difficulty-ordered exploration**: The agent preferentially practices tasks slightly beyond its current competence — maximizing learning per trial (the "zone of proximal development")

- **Competence tracking:** The agent monitors its own performance metrics and identifies skills that need practice
- **Self-testing:** The agent periodically re-executes previously learned tasks to verify retained competence — analogous to spaced repetition in human learning

## Applications

- **Long-deployment warehouse robot:** A warehouse robot deployed for months continuously learns new product appearances (A matrix updates), shelf layout changes (B matrix updates), and traffic pattern shifts (policy adaptation) — while maintaining proficiency on all previously learned tasks through concentration parameter protection.
- **Personal assistant robot:** A home robot learns its owner's preferences, routines, and household layout over weeks of cohabitation, building an increasingly personalized generative model that enables anticipatory assistance — "You usually have coffee at 7:30; I'll start the machine at 7:25."

## Conclusion

Lifelong learning enables autonomous robots to improve continuously throughout deployment. Self-supervised learning, forgetting mitigation, shift detection, and self-managed curricula implement Active Inference learning over extended timescales. The next module examines autonomous multi-agent communication.